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1 Exercise 1 – Identify Your Linux Filesystem

The storage configuration of the Linux system was examined using the following commands:

```
lsblk
lsblk -f
df -T
df -h
```

1.1 Storage Devices and Partitions

```
ananya@ubuntu-vm:~$ lsblk
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0       2:0    1     4K  0 disk
loop0     7:0    0     4K  1 loop /snap/bare/5
loop1     7:1    0  63.8M  1 loop /snap/core20/2599
loop2     7:2    0  63.8M  1 loop /snap/core20/2717
loop3     7:3    0  73.9M  1 loop /snap/core22/2010
loop4     7:4    0  73.9M  1 loop /snap/core22/2045
loop5     7:5    0 245.6M  1 loop /snap/firefox/6638
loop6     7:6    0 254.6M  1 loop /snap/firefox/7901
loop7     7:7    0 349.7M  1 loop /snap/gnome-3-38-2004/143
loop8     7:8    0 485.5M  1 loop /snap/gnome-42-2204/120
loop9     7:9    0   516M  1 loop /snap/gnome-42-2204/202
loop10    7:10   0  91.7M  1 loop /snap/gtk-common-themes/1535
loop11    7:11   0  12.2M  1 loop /snap/snap-store/1216
loop12    7:12   0  12.3M  1 loop /snap/snap-store/959
loop13    7:13   0  50.9M  1 loop /snap/snapd/24505
loop14    7:14   0  49.3M  1 loop /snap/snapd/24792
loop15    7:15   0   568K  1 loop /snap/snapd-desktop-integration/253
loop16    7:16   0   576K  1 loop /snap/snapd-desktop-integration/315
sda       8:0    0    40G  0 disk
├─sda1    8:1    0     1M  0 part
├─sda2    8:2    0   513M  0 part /boot/efi
└─sda3    8:3    0  39.5G  0 part /var/snap/firefox/common/host-hunspell
                                  /
sr0      11:0    1 155.3M  0 rom  /media/ananya/CDROM
sr1      11:1    1   4.7G  0 rom  /media/ananya/Ubuntu 22.04.3 LTS amd64
ananya@ubuntu-vm:~$
```

Figure 1: Storage devices and partitions identified using the lsblk command

The command `lsblk` displays the block devices connected to the system along with their partitions and hierarchy.

1.2 Filesystem Information

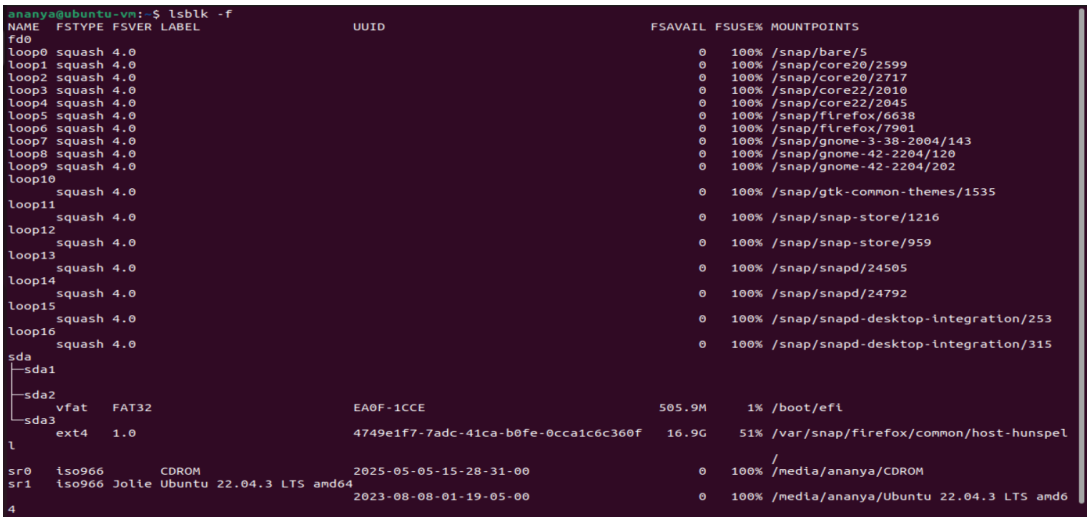


Figure 2: Filesystem information obtained using the lsblk -f command

The command `lsblk -f` displays filesystem types, labels, UUID information and mount points.

1.3 Filesystem Type

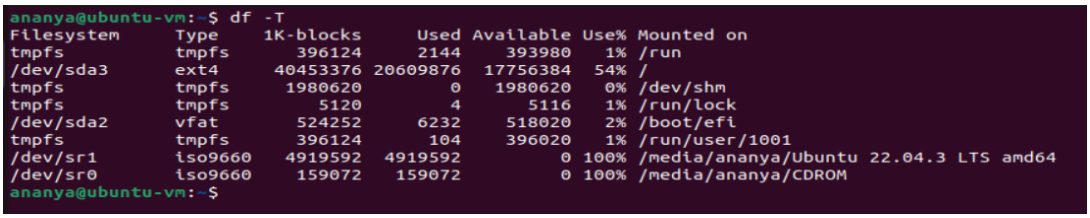


Figure 3: Filesystem type and mounted filesystems obtained using df -T

The command `df -T` was used to determine the filesystem type associated with mounted filesystems.

1.4 Storage Usage

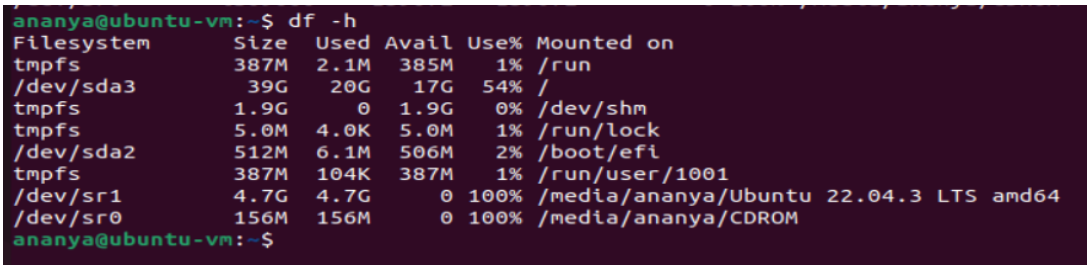


Figure 4: Filesystem capacity, used space and available space obtained using df -h

The command `df -h` displays filesystem usage in a human-readable format.

1.5 Observation Table

Device/Partition	Filesystem Type	Size	Mount Point
Refer to screenshot	Refer to screenshot	Refer to screenshot	Refer to screenshot

2 Exercise 2 – Disk → Partition → Filesystem → Mount Point

The storage organization of the Linux system was created based on the information obtained in Exercise 1.

```
ananya@ubuntu-vm:~$ lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
NAME      SIZE FSTYPE    MOUNTPOINTS
fd0         4K
loop0       4K squashfs  /snap/bare/5
loop1    63.8M squashfs  /snap/core20/2599
loop2    63.8M squashfs  /snap/core20/2717
loop3    73.9M squashfs  /snap/core22/2010
loop4    73.9M squashfs  /snap/core22/2045
loop5   245.6M squashfs  /snap/firefox/6638
loop6   254.6M squashfs  /snap/firefox/7901
loop7   349.7M squashfs  /snap/gnome-3-38-2004/143
loop8   485.5M squashfs  /snap/gnome-42-2204/120
loop9     516M squashfs  /snap/gnome-42-2204/202
loop10   91.7M squashfs  /snap/gtk-common-themes/1535
loop11   12.2M squashfs  /snap/snap-store/1216
loop12   12.3M squashfs  /snap/snap-store/959
loop13   50.9M squashfs  /snap/snapd/24505
loop14   49.3M squashfs  /snap/snapd/24792
loop15    568K squashfs  /snap/snapd-desktop-integration/253
loop16    576K squashfs  /snap/snapd-desktop-integration/315
sda        40G
├─sda1      1M
├─sda2    513M vfat      /boot/efi
└─sda3   39.5G ext4      /var/snap/firefox/common/host-hunspell
                         /
sr0     155.3M iso9660   /media/ananya/CDROM
sr1       4.7G iso9660   /media/ananya/Ubuntu 22.04.3 LTS amd64
ananya@ubuntu-vm:~$
```

Figure 5: Disk, partition, filesystem and mount point organization

A disk is a physical or virtual storage device that can contain one or more partitions. A partition can contain a filesystem, and the filesystem is attached to the Linux directory structure through a mount point. The root filesystem is mounted at `/`.

3 Exercise 3 – Explore the Root Filesystem

3.1 Current Working Directory

The current directory was determined using:

```
pwd
```

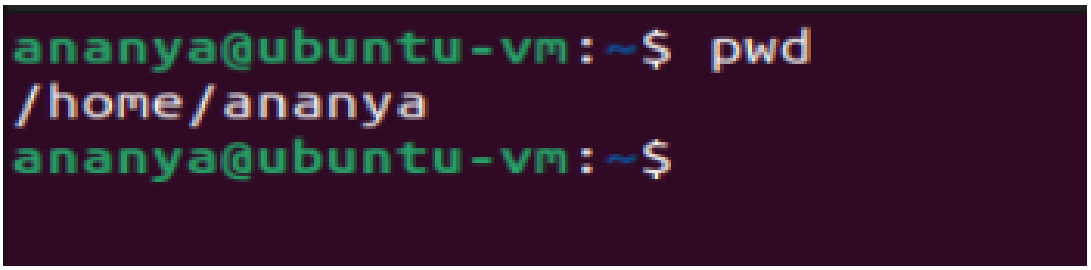


Figure 6: Current working directory obtained using pwd

3.2 Navigate to the Root Directory

The following command was used:

```
cd /
pwd
```

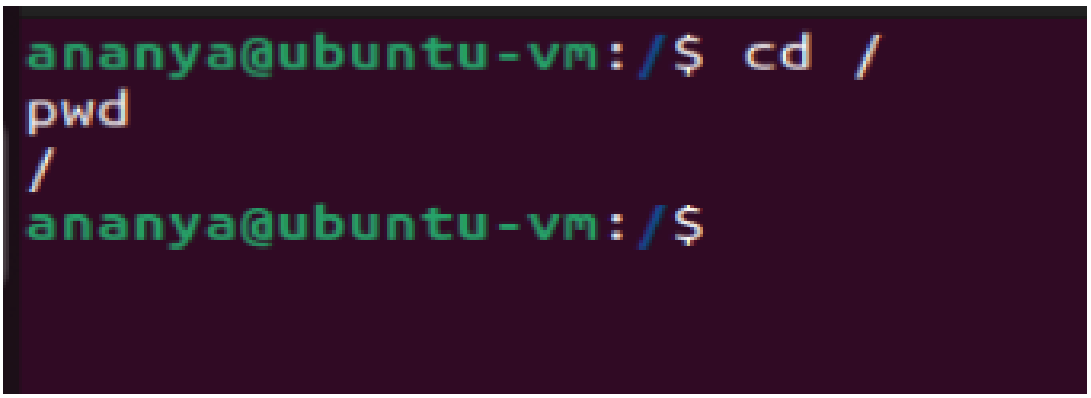


Figure 7: Navigation to the root directory

3.3 Contents of the Root Directory

The contents of the root directory were examined using:

```
ls
```

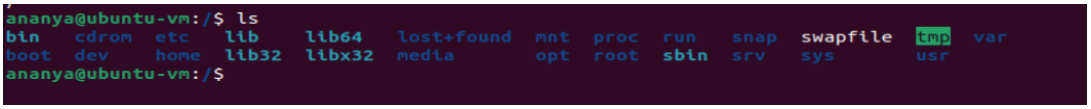


Figure 8: Contents of the root directory

3.4 Hidden Entries in the Root Directory

Hidden entries were examined using:

```
ls -a
```

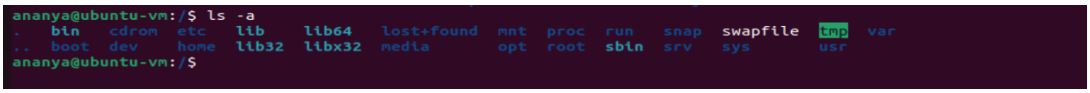


Figure 9: Hidden entries in the root directory

Directories directly available under the root directory include important locations such as `home`, `etc`, `var`, `usr`, `boot`, `dev`, `proc` and `tmp`.

4 Exercise 4 – Linux Filesystem Hierarchy Investigation

The following directories were investigated:

```
/home
/root
/etc
/var
/var/log
/tmp
/usr
/boot
/dev
/proc
```

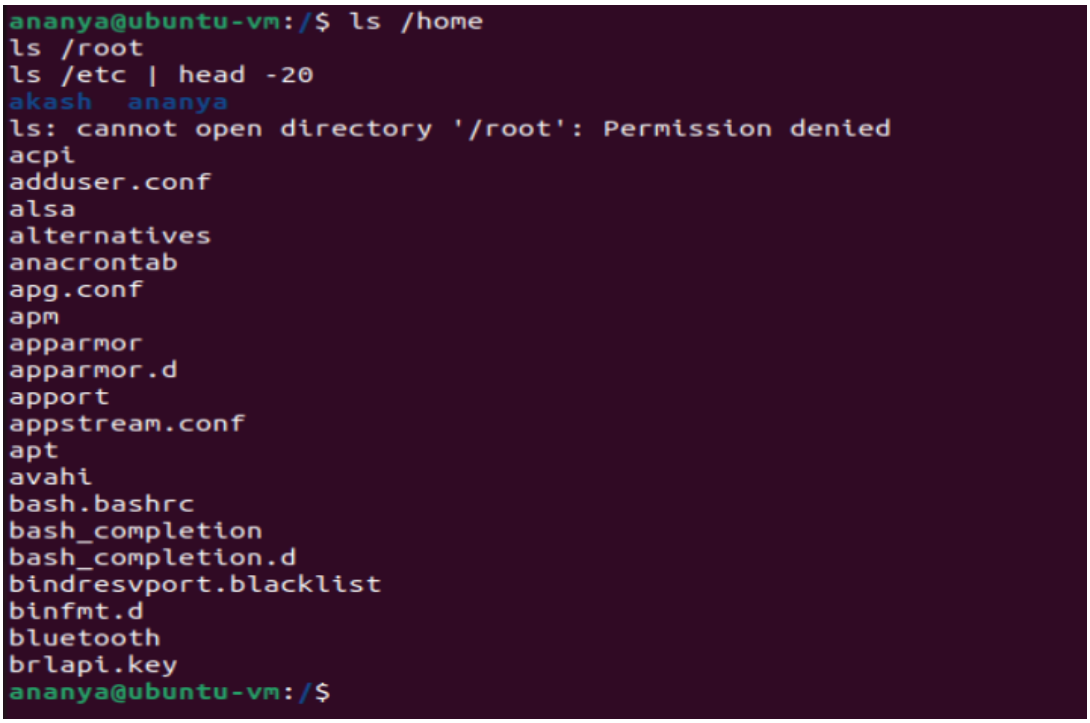


Figure 10: Contents of /home, /root and /etc

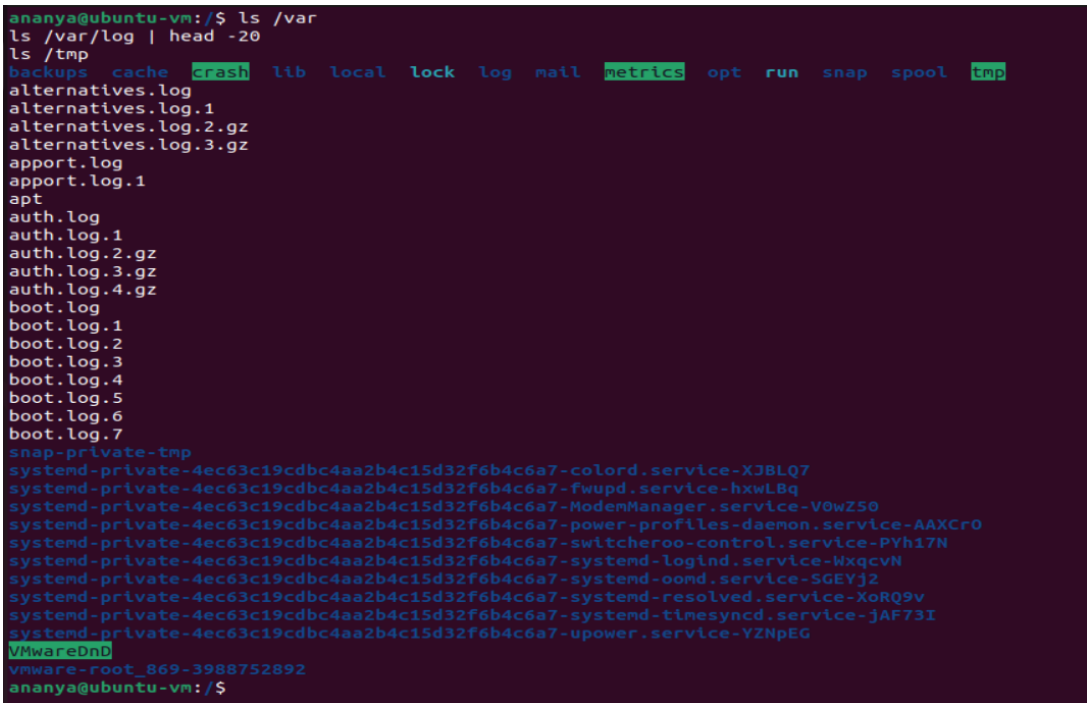


Figure 11: Contents of /var, /var/log and /tmp

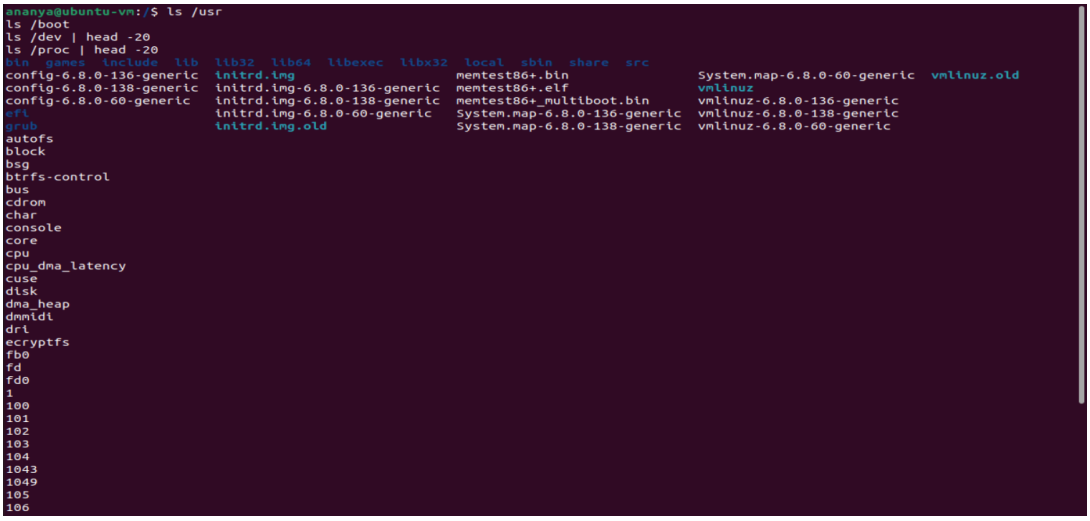


Figure 12: Contents of /usr, /boot, /dev and /proc

4.1 Observation Table

Directory	Example Contents	Purpose
`/home`	User directories and personal files	Stores files and directories belonging to normal users.
`/root`	Configuration and administrative files	Home directory of the root user.

/etc	passwd, group and hostname	Contains system-wide configuration files.
/var	log, cache and spool	Stores variable data generated by the system and applications.
/var/log	System and application log files	Stores logs used for monitoring, troubleshooting and investigation.
/tmp	Temporary files and directories	Stores temporary data created by users and applications.
/usr	bin, lib and share	Contains user programs, libraries and shared resources.
/boot	Kernel and boot-related files	Contains files required during the boot process.
/dev	null, zero and other device entries	Provides access to devices through filesystem entries.
/proc	CPU, memory and process information	Provides runtime information generated by the Linux kernel.

5 Exercise 5 – Find Your Home Directory

The username and home directory were identified using suitable Linux commands.

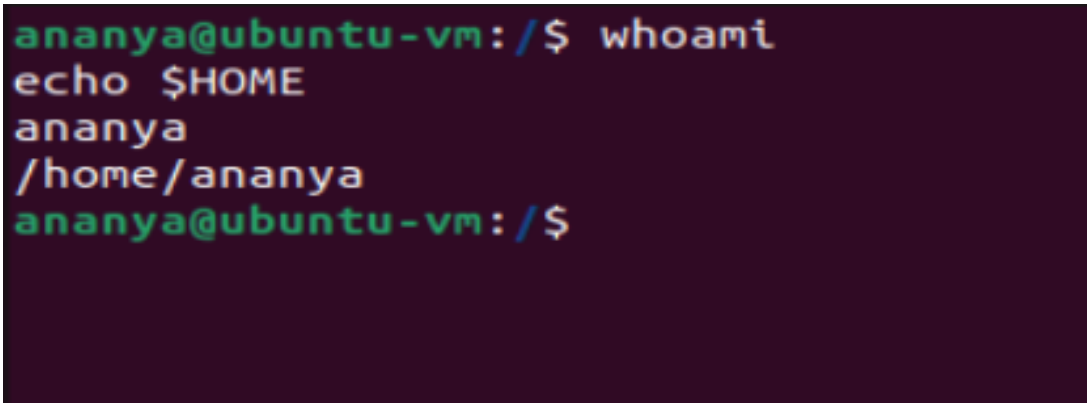


Figure 13: Username and home directory identification

The home directory was then accessed.

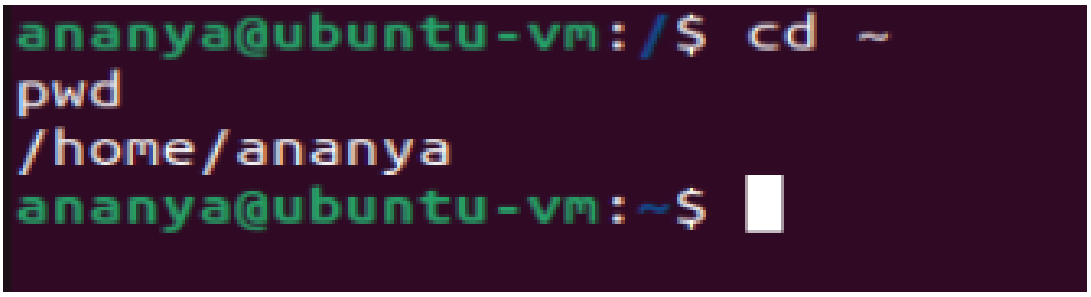


Figure 14: Navigation to the home directory

The normal contents were listed using:

```
ls
```



Figure 15: Normal contents of the home directory

All contents, including hidden entries, were listed using:

```
ls -a
```



Figure 16: All contents including hidden files in the home directory

6 Exercise 6 – Hidden Files Investigation

Linux identifies a hidden file or directory by a name beginning with a period (`.`).

A normal `ls` command does not display hidden files because they are intentionally omitted from the default listing.

The entry `.` represents the current directory, while `..` represents the parent directory.

A hidden file can still be accessed if its complete name is known and the user has the required permissions.

From a cybersecurity perspective, hidden files should not automatically be considered safe or malicious. Legitimate applications may use hidden files for configuration, while attackers may also attempt to hide suspicious files using names beginning with a period.

7 Exercise 7 – Build a Small Filesystem Hierarchy

The required `cyber_lab` directory hierarchy was created using Linux commands.

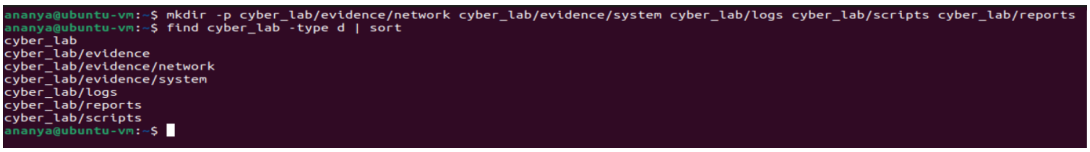


Figure 17: Creation of the cyber_lab directory hierarchy

The required files were created inside the corresponding directories.
The hierarchy was then verified.

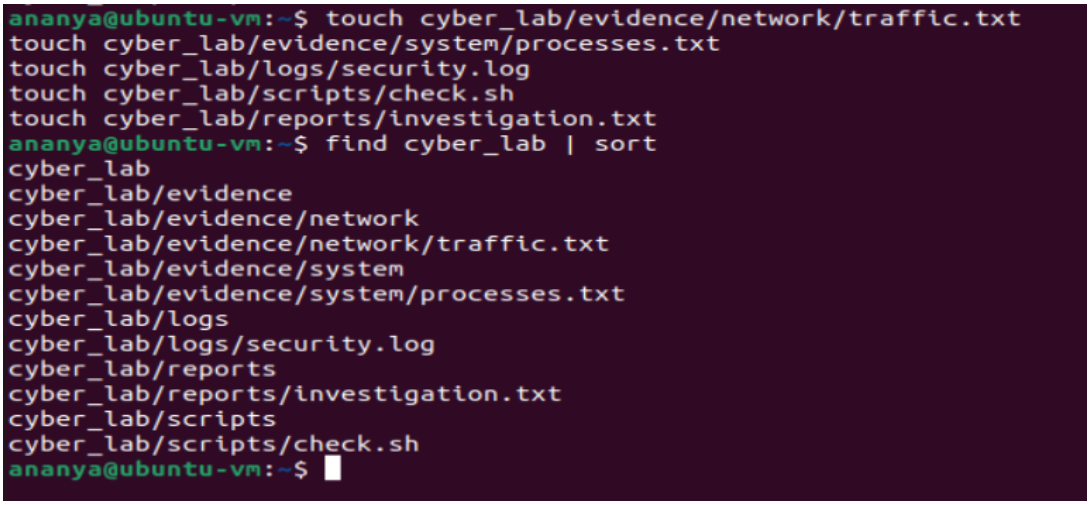


Figure 18: Complete cyber_lab directory hierarchy

The created structure contains the following important files:

- `cyber_lab/evidence/network/traffic.txt`
- `cyber_lab/evidence/system/processes.txt`
- `cyber_lab/logs/security.log`
- `cyber_lab/scripts/check.sh`
- `cyber_lab/reports/investigation.txt`

8 Exercise 8 – Absolute and Relative Paths

The current directory was entered as `cyber_lab`.

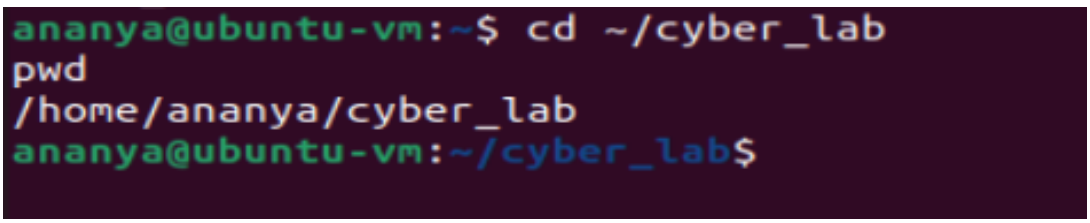


Figure 19: Entering the cyber_lab directory

8.1 Relative Path Navigation

The `evidence/network` directory was accessed using a relative path.

```
ananya@ubuntu-vm:~/cyber_lab$ cd evidence/network
pwd
/home/ananya/cyber_lab/evidence/network
ananya@ubuntu-vm:~/cyber_lab/evidence/network$
```

Figure 20: Navigation using a relative path

8.2 Absolute Path Navigation

The `/etc` directory was accessed using an absolute path.

```
ananya@ubuntu-vm:~/cyber_lab/evidence/network$ cd /etc
pwd
/etc
ananya@ubuntu-vm:/etc$
```

Figure 21: Navigation to /etc using an absolute path

8.3 Return to Home Directory

The home directory was accessed using the shortest possible command.

```
ananya@ubuntu-vm:/etc$ cd
pwd
/home/ananya
ananya@ubuntu-vm:~$
```

Figure 22: Return to the home directory

8.4 Navigate to Reports Directory

```
ananya@ubuntu-vm:~$ cd cyber_lab/reports
pwd
/home/ananya/cyber_lab/reports
ananya@ubuntu-vm:~/cyber_lab/reports$
```

Figure 23: Navigation to the reports directory

8.5 Move One Directory Upward

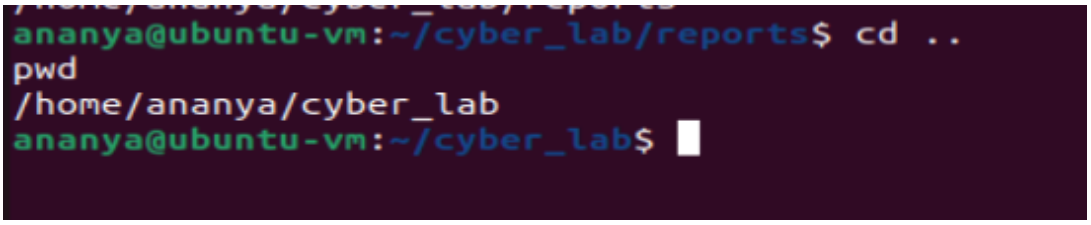


Figure 24: Moving one directory upward

An absolute path begins from the root directory `/`. A relative path is interpreted from the current working directory.

The symbol `/` represents the root of the filesystem hierarchy, while `~` represents the current user's home directory.

9 Exercise 9 – Identify File Types

System objects and files created in the previous exercise were examined using:

```
file <filename>
ls -l <filename>
```



Figure 25: File types of selected system objects

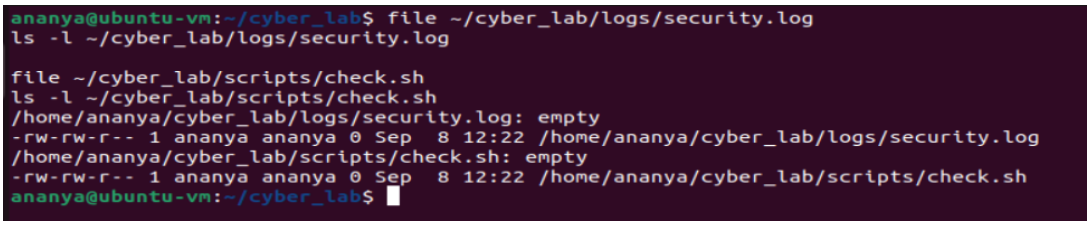


Figure 26: File types of files created inside cyber_lab

Object	Type	
/etc/passwd	Regular text file containing user account information	
/etc	Directory	
/bin/bash	Executable program file	
/dev/null	Character device file	
/proc	Virtual filesystem directory	
security.log	Regular file	
check.sh	Regular script file	

10 Exercise 10 – Examine File Metadata

The file `investigation.txt` was updated with a small amount of text.

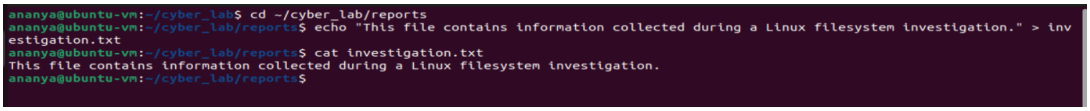


Figure 27: Content added to investigation.txt

The file metadata was examined using:

```
ls -l
stat investigation.txt
```

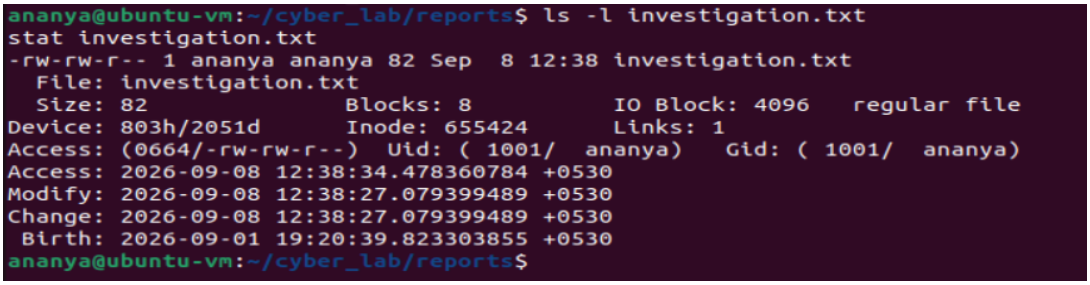


Figure 28: Metadata of investigation.txt

File metadata includes information such as filename, size, type, owner, group, inode number and timestamps.

Metadata is not the same as file content. File content is the actual data stored in the file, whereas metadata describes properties of that data.

When file contents are modified, the modification time is updated.

11 Exercise 11 – Explore /etc

Important files inside `/etc` were located.

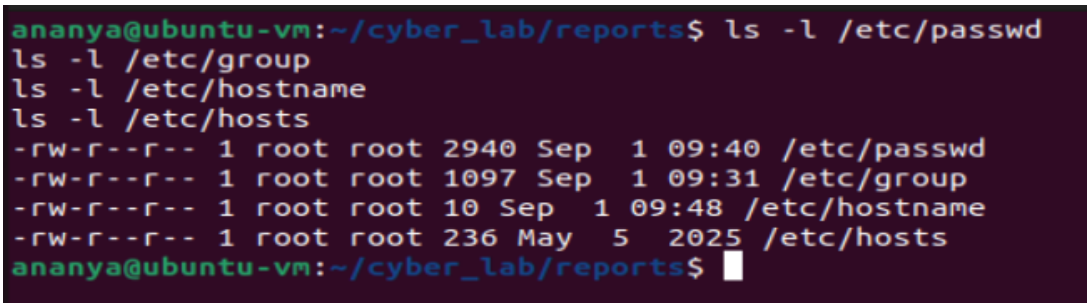


Figure 29: Important files located inside /etc

The contents of permitted files were examined.

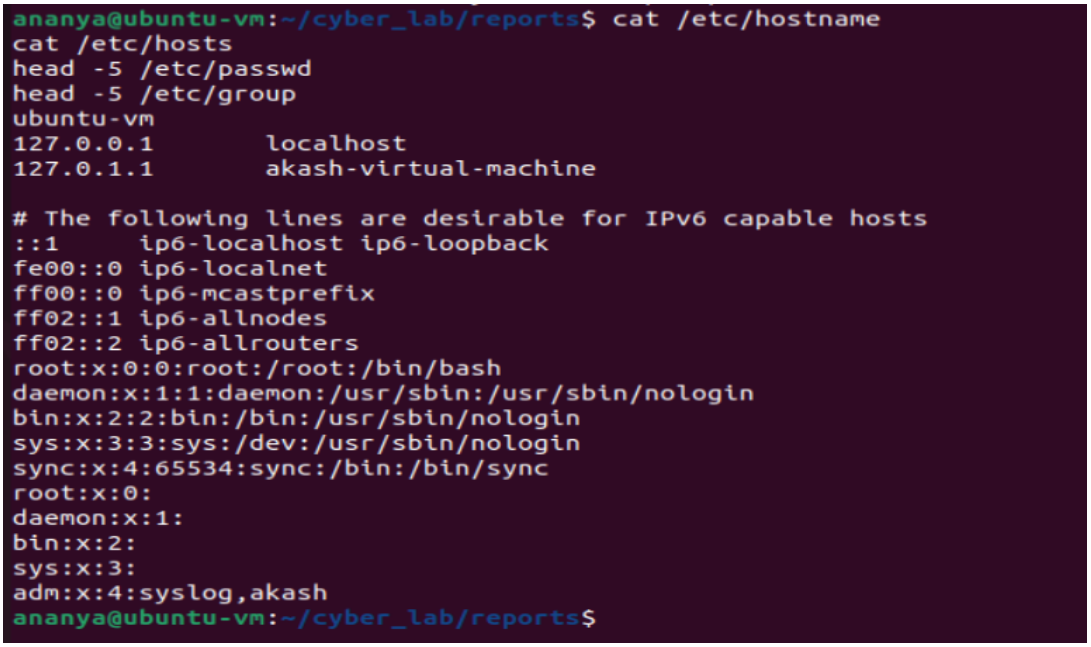


Figure 30: Contents of selected files inside /etc

The `/etc` directory primarily stores system-wide configuration information.

The file `/etc/passwd` contains user account information.

The file `/etc/hostname` contains the hostname of the system.

The `/etc` directory is security-sensitive because changes to configuration files can affect authentication, networking, services and overall system behavior.

12 Exercise 12 – Explore Linux Log Storage

The contents of `/var/log` were investigated.

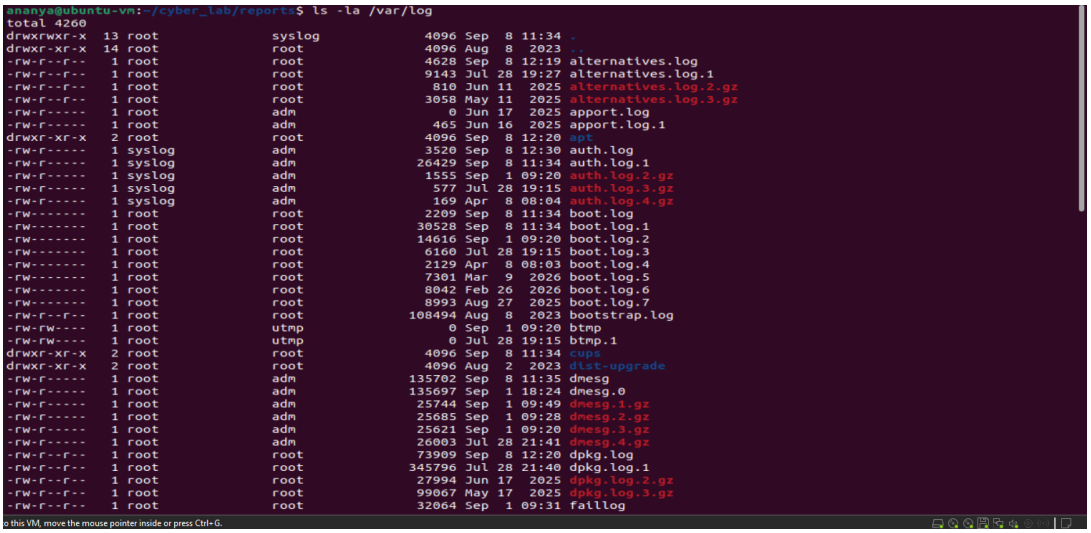


Figure 31: Available files and directories inside /var/log

Recently modified entries were identified.

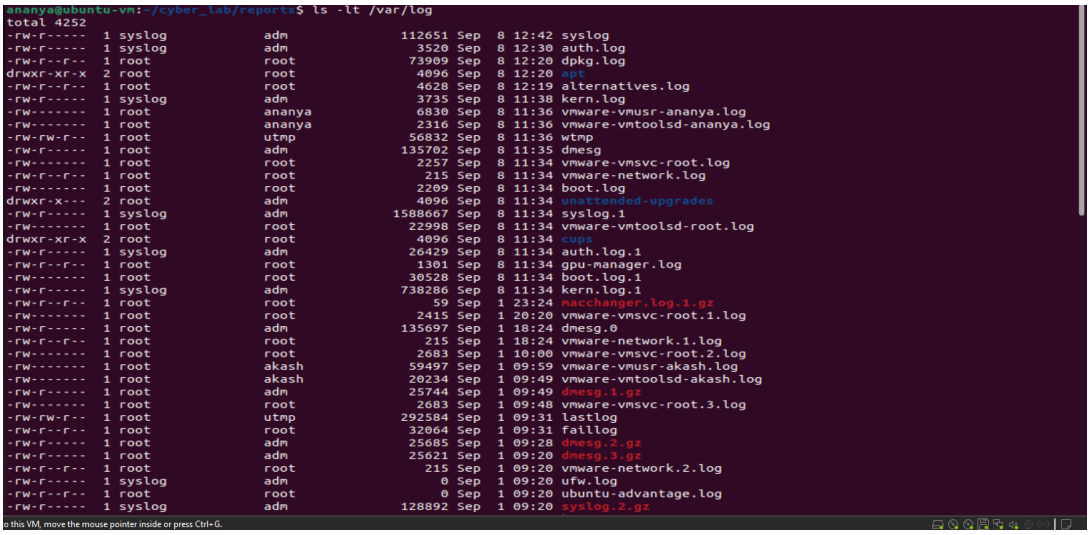


Figure 32: Recently modified entries in /var/log

Readable logs were examined.

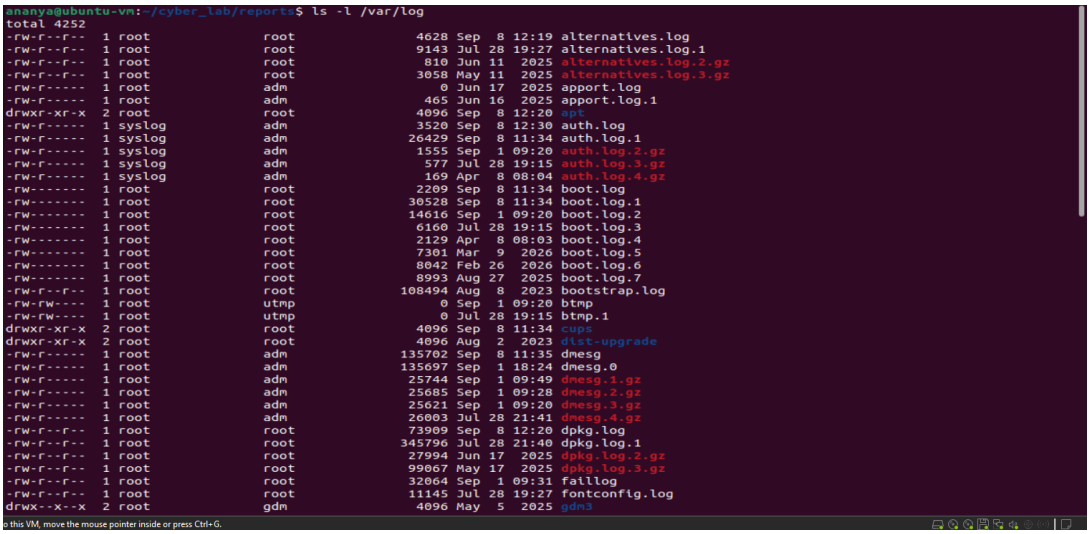


Figure 33: Contents of a readable log file

An attempt was made to access a restricted log.

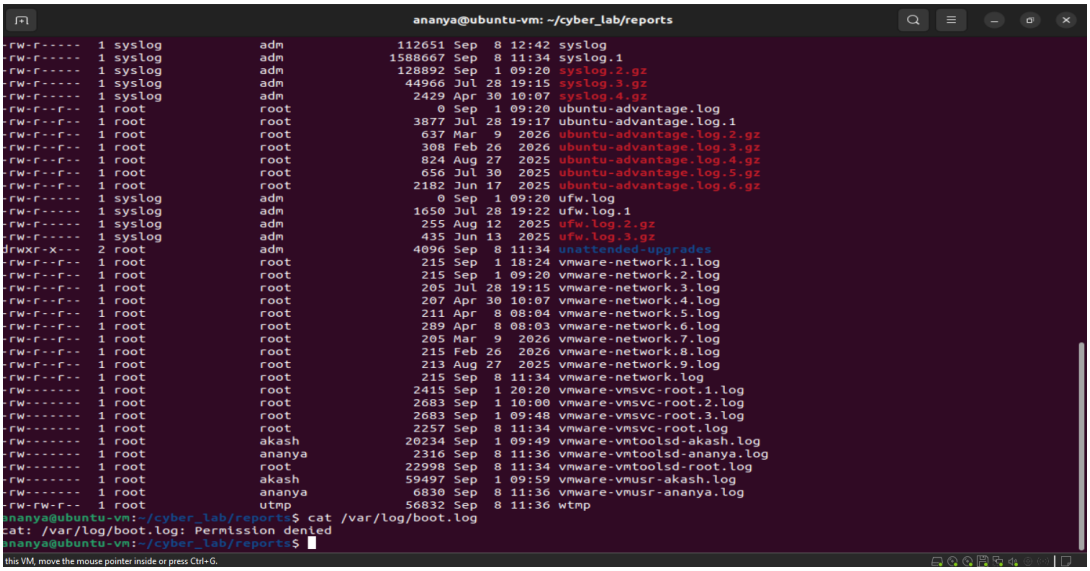


Figure 34: Restricted log access as a normal user

Linux maintains logs to record system events, service activity, authentication events and errors.

Logs are useful during incident investigations because they can help reconstruct events and identify suspicious activity.

Some logs are restricted because they may contain sensitive system or security information.

13 Exercise 13 – Investigate /tmp

The temporary directory was examined without modifying files belonging to other users.

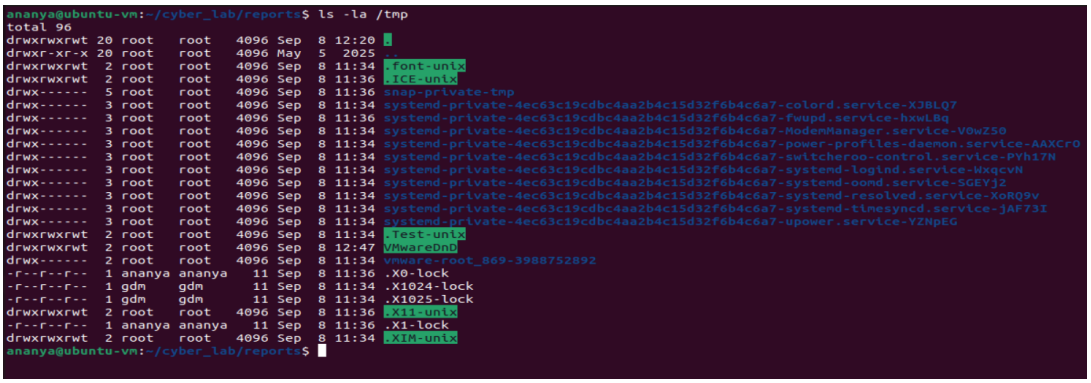


Figure 35: Contents of /tmp

Hidden entries were checked.

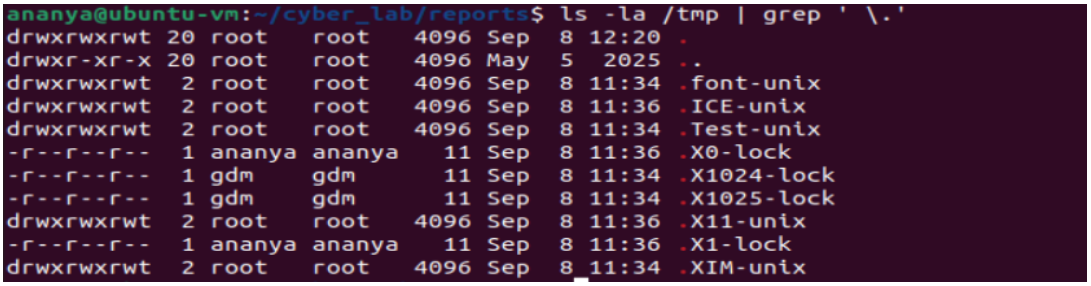


Figure 36: Hidden entries inside /tmp

Permissions and ownership were examined.

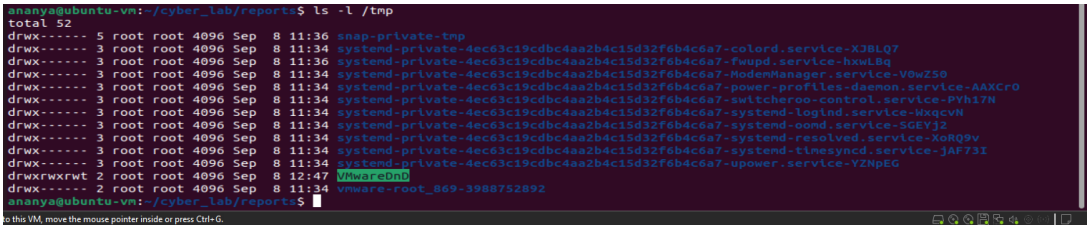


Figure 37: Permissions and ownership of entries inside /tmp

Recently modified entries were identified.

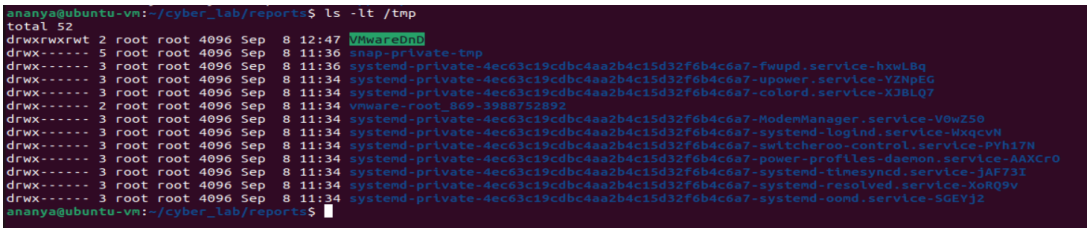


Figure 38: Recently modified entries inside /tmp

Temporary directories can be relevant during cybersecurity investigations because malicious programs or scripts may store temporary files, downloaded payloads or traces of suspicious activity in such locations.

14 Exercise 14 – Explore the /proc Filesystem

CPU information was examined.



Figure 39: CPU information obtained from /proc/cpuinfo

Memory information was examined.

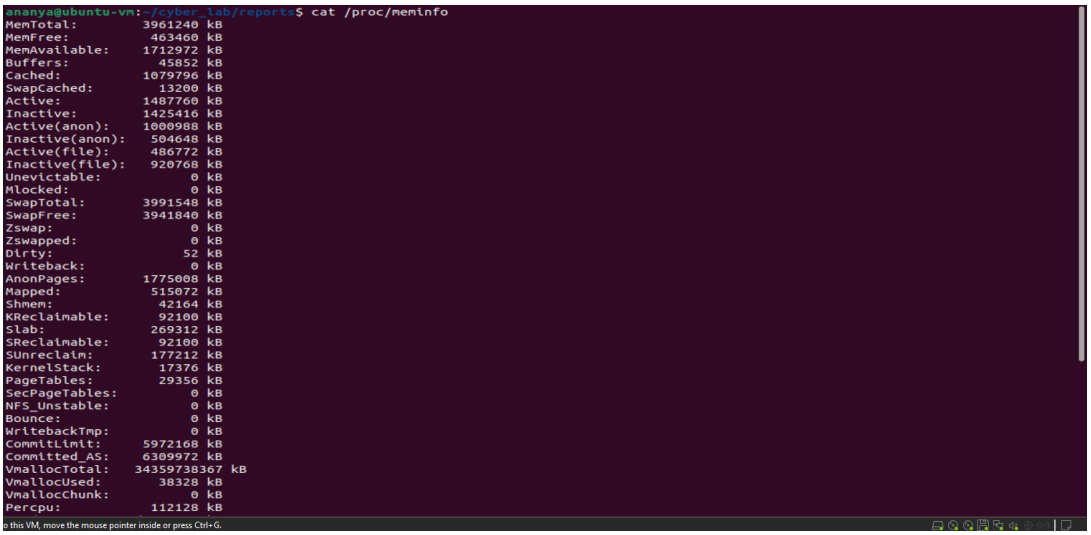


Figure 40: Memory information obtained from /proc/meminfo

The current shell PID was identified.

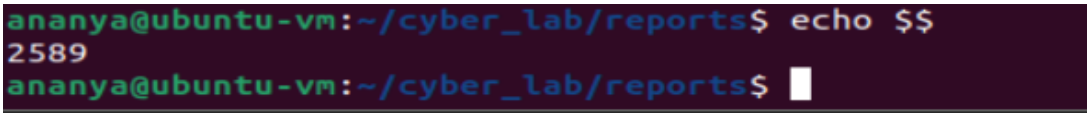


Figure 41: Current shell process identifier

The corresponding directory was checked inside `/proc`.

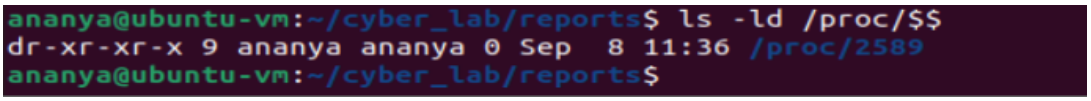


Figure 42: Directory corresponding to the current shell PID inside /proc

The file `/proc/cpuinfo` provides information about processors available to the system.

The file `/proc/meminfo` provides information about memory usage and availability.

Numbered directories inside `/proc` represent running processes and their process identifiers.

The `/proc` filesystem is not an ordinary disk-based filesystem. It is a virtual filesystem generated dynamically by the Linux kernel.

It is useful during security investigations because it provides information about processes, system configuration and runtime activity.

15 Exercise 15 – Explore /dev

The `/dev/null` device was examined.

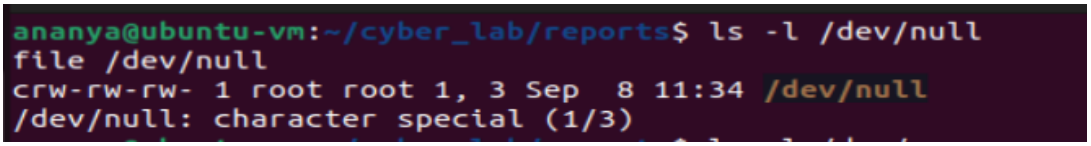


Figure 43: Information about /dev/null

The `/dev/zero` device was examined.

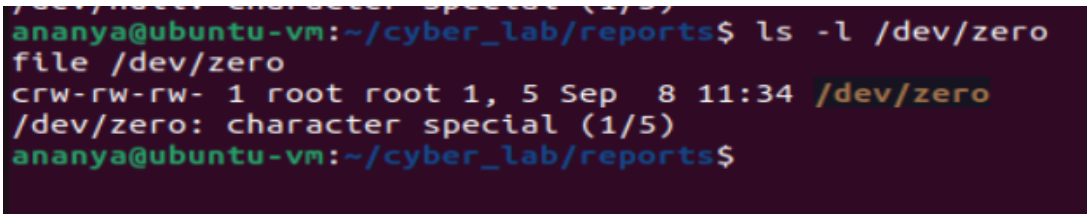


Figure 44: Information about /dev/zero

The `/dev` directory represents device files through which Linux applications interact with hardware and virtual devices.

The entries are not ordinary user files. They represent device interfaces.

The file `/dev/null` is a character device.

Linux exposes devices through the filesystem to provide a consistent interface for accessing different devices.

16 Exercise 16 – Filesystem Space Investigation

The space occupied by the home directory was examined.

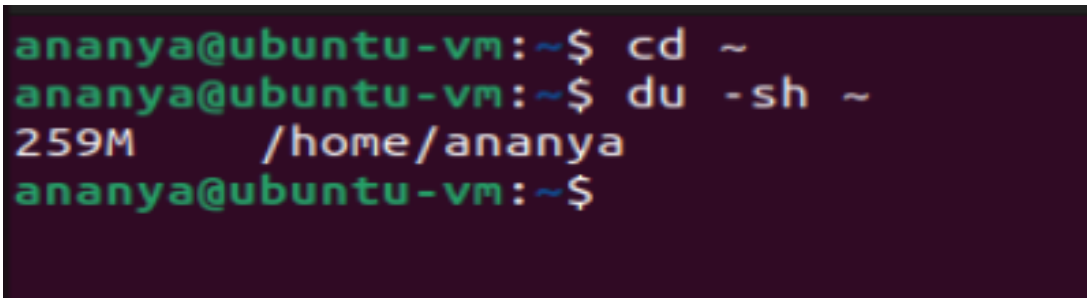


Figure 45: Disk usage of the home directory

The space occupied by the `cyber_lab` directory was examined.

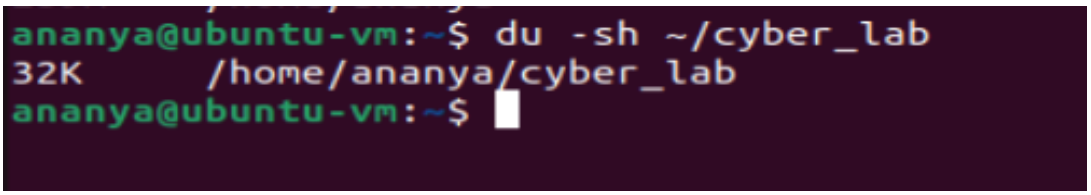


Figure 46: Disk usage of the cyber_lab directory

The command `df` reports disk space usage for mounted filesystems.

The command `du` reports the amount of disk space occupied by files and directories.

Therefore, `df` provides filesystem-level usage information, whereas `du` provides directory and file-level usage information.

17 Exercise 17 – Cybersecurity Investigation Scenario

During a cybersecurity investigation involving suspicious activity, the filesystem locations below would be important.

Location	**Why Investigate**	**Security-Relevant Information**
`/home`	Contains user files and personal data.	Documents, downloads, scripts and potentially suspicious user files.
`/etc`	Contains important system configuration.	User account information, service configuration, networking and authentication settings.
`/var/log`	Contains system and application logs.	Authentication events, service activity, errors and timestamps of suspicious events.
`/tmp`	Temporary files may be created by applications or malicious software.	Temporary scripts, suspicious downloads and short-lived artifacts.

`/proc`	Provides live information about the running system.	Running processes, memory information and process identifiers.
`/dev`	Contains device interfaces.	Information about available devices and unusual device-related activity.
`/root`	Contains files belonging to the root user.	Administrative scripts, configuration files and privileged user data.